Gullak: live the money life you didn't live

An adaptive, multilingual financial-literacy platform where Grade 6–12 students live through India's real money moments with play money, and see what every choice would have changed.

1. Problem statement and our understanding

Vanguard asks for real financial literacy for school students in Grades 6–12, especially in underserved Hyderabad communities, across four pillars: **plan & budget**, **save & manage**, **understand investing** and **stay financially safe**, learnt by doing. What we see:

- **Consequences are invisible.** Money decisions pay off or hurt years later, so students never feel the link between a choice and its outcome. Existing finance games teach rules; none let you see the alternative.
- **One size does not fit.** A Class 6 and a Class 12 student need different depth, and students differ in format (story, visual, formula, simulation) and language (English, Hindi, Telugu).
- **Real conditions:** cheap Android phones, slow or no internet, shared devices, facilitator-led classrooms.
- **Trust:** students must never be pushed to real products, and must learn to recognise scams aimed at them.

2. Idea: what makes Gullak new

Core loop: learn it, unlock it, invest in it. Every student gets a virtual starting balance and their own paper portfolio on **real Indian market prices**. Study a topic (savings, FD, RD, gold, mutual funds, index funds, stocks), pass its quiz, and that instrument **unlocks**: open a virtual FD, start an SIP, buy index-fund units or a stock at today's real price, and watch it move. Students effectively run their own market from Class 6 to 12, piggy bank first and stocks last, so every investment is something they've just learnt.

Money Multiverse: see the life you didn't live

Every big decision splits the student's life. They keep living their path while a **parallel self** lives the other choice, on **real Indian market history**. Saved or spent in 2019? Broke the FD, spent the emergency fund or sold the index fund when a ₹15,000 hospital bill hit in March 2020? Both lives run on real prices and can be opened side by side at any time. The takeaway is never "time the market"; it's "nobody knew tomorrow, and the emergency fund meant you didn't have to choose".

Live India's real money moments

Practice episodes replay real events with real data: demonetisation night (2016), the March 2020 crash, gold in 2020, the 2022 rate rises. Life events (a fee due, a phone breaks, a job loss at home) force real trade-offs. Then **Live** mode paper-trades today's prices, where tomorrow is unknown, scored on habits, never returns.

Play the other side

Be the bank: run a village bank, take deposits, lend, set rates and survive a bank run, so interest finally makes sense.

Think like a scammer: design a fake "double your money" scheme in a sandbox and watch who falls for it; building the red flags teaches them.

Scam drills, like fire drills

Without warning, a realistic fake SMS, WhatsApp or UPI "collect" request appears. The student must spot it and report it (1930, cybercrime.gov.in), then gets a debrief. Practising the moment, not reading about it.

Teach the AI, take it home

Students correct a confused AI classmate's money mistakes: learning by teaching. Lessons come home in Telugu or Hindi audio so children teach parents about UPI safety, gold and chit funds.

Underneath sits a 33-topic curriculum across the four pillars, as a prerequisite graph tagged by grade group (6–7, 8–10, 11–12). Each topic is stored once and shown in the format that suits the learner (illustrated comics and stories, visual explainers with real-data charts, short videos, formulas with live calculators, audio, simulations and quizzes), plus quick games (need or want, scam or real, Budget Bazaar) and branching comic stories with a recurring cast who grow up from Class 6 to 12. Every answer, game and simulation decision is evidence: **knowledge tracing** estimates mastery per concept and a **format bandit** learns how each student learns best.

Evidence → **Recommendation** → **Activity** → **Simulation** → **Mastery update** → **Changed recommendation**

Progress feels good without being about money: a piggy bank that fills, saving streaks, badges for *habits* ("kept an emergency fund", "spotted 10 scams"), instruments that unlock like levels, and class challenges for facilitators. There are no leaderboards for money or returns.

Hard guardrail. Gullak educates; it never recommends a real product or tells anyone to buy or sell. All money is virtual, and the type system cannot represent real money. Every projection shows its assumptions, source and as-of date. The AI tutor screens questions and answers for advice.

3. High-level approach and architecture

1. **Onboard:** language → grade group → current experience (only a starting prior) → optional format preference ("Mix it up" by default).

2. **Recommend:** topics open when prerequisites are mastered and they suit the student's grade group; passing unlocks instruments, games and episodes.
3. **Learn:** lessons pre-generated once per topic × format × grade group × language, reviewed and cached, so they're free at run time and work offline.
4. **Play:** games, stories and the Multiverse simulation on real data, with source and as-of dates shown.
5. **Adapt:** evidence updates mastery and format preferences, which change the next recommendation, with a "Why this?" explanation.

Responsive PWA (Next.js, React, TypeScript)
phone + desktop · installable · offline lessons · English / Hindi / Telugu (Sarvam translation, human review)

Engine (pure TypeScript) FD/RD/SIP/compounding maths, early-exit rules, valuation, unlock rules, sim engine with counterfactual branches; unit-tested	**Content as data** topics, graph, games, stories, episodes and event decks as JSON validated by Zod; new topics, games or languages need no code	**Adaptive models** knowledge tracing trained in Colab, exported as JSON and run in the app; format bandit online
Real Indian data AMFI NAVs, NSE indices, MoSPI CPI, IBJA gold, bank and small-savings rates, RBI, NPCI; cached adapters with snapshot fallback	**Backend (Supabase / Postgres)** one data-access layer, row-level security, server-graded quizzes, atomic idempotent virtual transactions	**AI tutor + classmate** provider-agnostic (Gemini → Groq → content search), versioned prompts, input/output guardrails, rate limit, cache

A facilitator dashboard shows class progress and weakest concepts and assigns activities. A stretch goal is an Account Aggregator demo on clearly labelled synthetic data, turning spending patterns into learning nudges.

4. Planned testing methodology

Layer	What we test and how
Finance engine	Unit tests for every formula, rule and branch with normal, edge and invalid inputs; integer paise (no float drift); checked against hand calculations and published FD/RD maturity values. Parallel branches must start from identical states and differ only by the choice.
Content and data	Every topic, game, story and episode is schema-validated in CI. The graph must have no cycles or unknown prerequisites. Episodes must be covered by real committed price series on their event and reveal dates. Stories must have no dead ends. All locales must have identical keys.
Backend and safety	Isolated Postgres tests for row-level security, idempotent retries, concurrency and rollback. A red-team set of advice-seeking questions (English, Hindi, Telugu, Hinglish) must be refused and redirected to teaching, and model output is screened for product names and buy/sell language.
End to end	Playwright runs onboarding → lesson → quiz → unlock → simulation on phone and desktop in CI on every push; a performance budget on a throttled low-end Android profile; offline lesson checks.
Learning impact	Simulated learners check that evidence changes the next recommendation. Short sessions with students: pre/post quiz per pillar, game replay rates, scam-drill detection rate, and whether seeing the parallel life changes their emergency-fund choice.